12. Kinetic Voltmeter

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We explore dropping of alkaline batteries as a simple way to detect the state of charge. Literature on this topic generally does not delve into the mechanical aspects of analysing the bounces, so a mechanical approach inspired by the underlying electrochemistry is sought. Another non-invasive approach based on acoustic principles is suggested.

I. INTRODUCTION

The "battery bounce" test is the simplest way to figure out if a battery is discharged or not. The claim is that dead batteries bounce to higher heights than fresh batteries. Therefore, the aim of this study is to simply verify this, and to provide another non-invasive mechanical method to estimate the charge of a battery.

For zinc-manganese oxide alkaline batteries like the ones we discuss here, Hall et al. [1] reason that since the governing chemical reaction proceeds as

$$\mathrm{Zn\,(s) + 2\,MnO_2(s) + H_2O\,(l) \longrightarrow ZnO\,(s) + 2\,MnOOH\,(s)}$$

and since the bulk modulus of Zn increases from 59 to 134 GPa [2] [3] as it transforms to ZnO, the formation of ZnO must be the cause of the higher bounce. In fact it is well known that ZnO is added to materials to increase their bouncing properties. To quantify the bounce we use of course, the coefficient of restitution (COR) e and seek its variation as a function of the state of charge (SOC). As per the claims then, we expect a monotonically decreasing behaviour.

We begin by describing the details of our experimental setup, and then go on to explain a simple macroscopic model that explains the contact dynamics of a cylinder colliding against an infinite half-space. We then elaborate on a detailed microscopic model that qualitatively explains features of Figure 2. After this, we elaborate on acoustic methods of estimating SOC.

II. ANALYSIS

A. Experimental Setup

Two sets of experiments were conducted to investigate the correlation between the COR and the SOC of primary AAA zinc manganese alkaline cells. In the first phase, the battery was discharged under controlled conditions using a custom-built electronic load. The discharge current was determined from the output voltage according to

$$I(t) = \frac{V_{battery}(t)}{R_1 + R_2} \approx \frac{V_{battery}(t)}{R_1}$$

where R_2 is a small shunt resistor as in Figure 1. The instantaneous current was integrated in real time to estimate the discharged capacity. At increments of 100 mAh, the open-circuit voltage of the cell was recorded, followed by drop tests performed using a vertically aligned tube apparatus designed to ensure consistent impact orientation (see Figure 3). The drop experiments were carried out sequentially on three surfaces of differing rigidity: a wooden lecture table **LT**, a concrete floor **CF**, and a hard-cover book **HC**. Drops were conducted from heights of 12 cm, 15 cm, and 20 cm, respectively. Each trial was repeated thrice to ensure reproducibility, and impact heights were measured with a precision scale. Videos were then intensively tracked using Tracker [4].

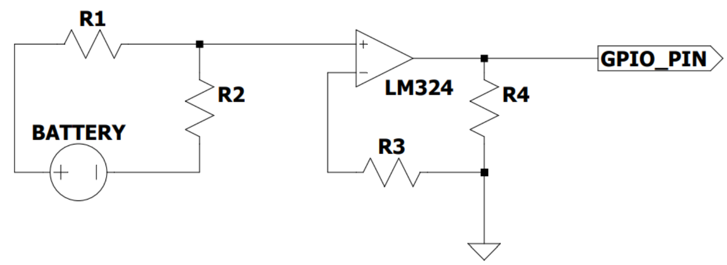

FIG. 1: In-house electronic apparatus built for regulated battery discharge

In the subsequent phase, additional experiments were performed employing an updated drop apparatus on a rubber surface **RS** and a tiled floor **TF**, for multiple launch heights. This extended dataset enabled comparison across both rigid and compliant impact conditions, providing further insight into the restitution behavior as a function of SOC and surface characteristics. Dropping from three different heights and taking mean COR for the surface **LT**, we obtain Figure 2.

TABLE I: Experimental Components

Object	**Specification**
Battery	Duracell AAA Alkaline, 1.5 V
R_1	3.9 Ω, 5% tolerance, 0.5W ceramic resistor
Shunt Resistor R_2	75 $m\Omega$, 5% tolerance
Microcontroller	ESP32-WROOM-32, 240 MHz
DAQ	NI USB-6009, 14-bit
3D Vertical Stand	24 cm height
Phone Camera	480 FPS

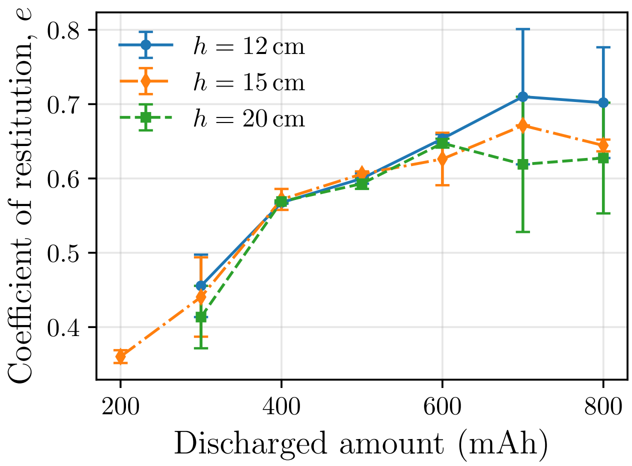

FIG. 2: Discharged Batteries Bounce Higher

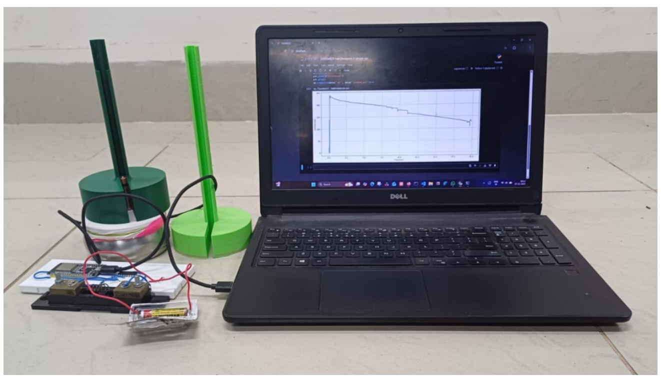

FIG. 3: Experimental Setup

B. Contact Dynamics

Let the battery be modelled as a rigid cylinder dropped from a height h_0, then it hits the ground at time τ where

$$\tau = \sqrt{\frac{2h_0}{g}}.$$

where g is the acceleration due to gravity. There exist many ways to model the contact phase. A simple choice is the so-called Kelvin-Voigt element, which is nothing but a damped harmonic oscillator. In such models, the collision force is modelled as

$$F_{collision} = -k\delta - c\dot{\delta}$$

with δ being the relative compression between surfaces. The spring element represents the tendency to bounce back after collision, and the damping force represents loss in energy. Such a model has the advantage of simplicity, but has two critical disadvantages:

- The coefficient of restitution is *independent* of initial velocity.
- Energy losses scale as the *square* of initial velocity rather than the *cube*, as is experimentally observed [5].

The second point follows from the first since

$$v_f = ev_i \implies \Delta E = \frac{1}{2}mv_i^2 - \frac{1}{2}mv_f^2 = \frac{1}{2}mv_i^2(1-e^2)$$

with v_f and v_i being the final/initial velocities just after/before the collision. The first point is proved in the Supplementary Information.

Experimental observations indicate that the coefficient of restitution (COR) e actually varies with small enough v_i as

$$e = 1 - \alpha v_i$$

This would explain the cubic dependence of loss of energy on v_i

$$\Delta E = \frac{1}{2}mv_i^2(1-e^2) = \frac{1}{2}mv_i^2(1-(1-\alpha v_i)^2) \approx \alpha m v_i^3$$

To explain the empirical linear dependence of e on v_i, a more realistic model was suggested by Hunt and Crossley [6], where the collision force has the functional form

$$F_{collision} = -k\delta^n - \lambda\delta^n\dot{\delta}$$

where it was shown that $\lambda = \frac{3}{2}\alpha k$. It can also be shown that our system of a cylindrical battery impacting a flat half-space corresponds to $n = 1$ [7]. For the problem at hand, we shall soon relate these model parameters to the charge of the battery. Let us now see some implications of such a choice of damping.

We numerically solve the above ODE to model the collision using a semi-implicit Euler scheme with a time step $dt = 10^{-7}$, which is much smaller than the typical time scale of the problem. It must be noted that for the Hunt-Crossley model the time of impact was found to be very slowly varying with λ. This is important because given the time of impact T from precise slow motion frame-by-frame analysis, we can estimate k as $k \sim \pi^2 m/T^2$, where m is the mass of the battery. The overall logical flow of model validation is hence described below.

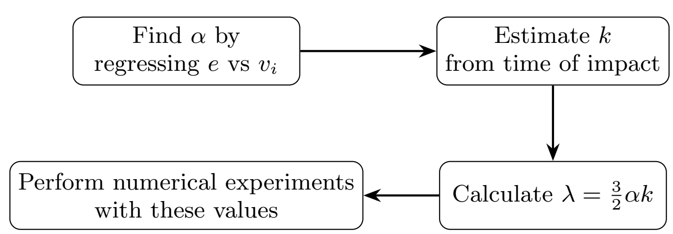

FIG. 4: Model Validation Flowchart

For the particular case of dropping on **RS**, with a half charged battery (400 mAh), we found $\alpha \approx 0.0178$, and a time of impact 8.01 ± 0.01 ms. Following the workflow highlighted above, we obtain Figure 5.

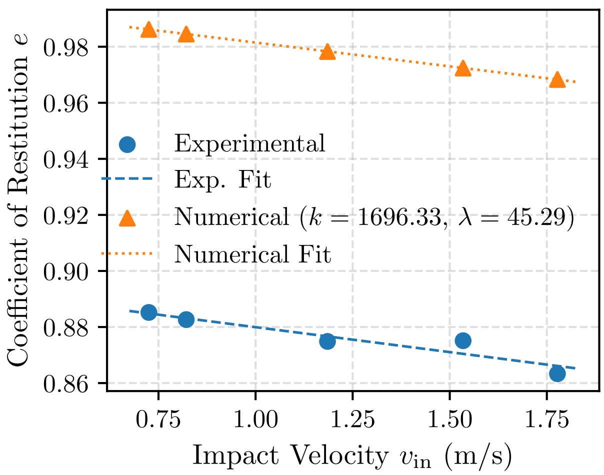

FIG. 5: Numerical-Experimental Comparison for the Hunt-Crossley Model

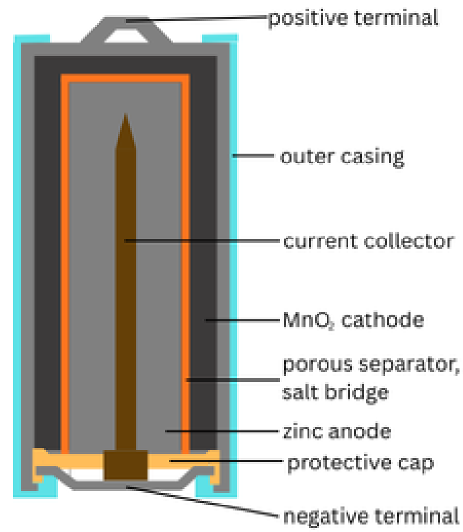

FIG. 6: Cylindrical Alkaline Battery Cross Section

One can even track the maximum height after each bounce, and analytically it can be shown (refer to Supplementary Information) that this follows the recursion

$$h_{n+1} = (1 - \alpha\sqrt{2gh_n})^2 h_n$$

Fitting this numerically can give α, but we postpone this discussion to subsection IIC where we discuss a more precise recursion.

The constant error observed that creeps in could be a) systematic: finite resolution of motion in the tracked videos, improper drop of batteries, etc. and b) since we have ignored the effects of viscous forces on the motion of the battery. This can only be ignored when the battery is discharged and the material inside is rigid. To truly understand how COR varies with SOC, we must think microscopically.

C. Microscopic Explanation

The energy loss in a bouncing battery cannot be explained by treating it as a simple elastic cylinder. Two dissipation mechanisms are involved. The first is impact-related loss (sound, vibration, and heat) already captured by the previous model. The second is viscous dissipation inside the anode gel, arising from relative motion between Zn/ZnO particles and the gel matrix; this occurs only while these particles remain mechanically unbound from the rest of the cell.

Studies [8] show that ZnO formed during discharge creates bridges between zinc particles, generating an increasingly connected internal network. This network grows radially inward from the separator toward the collector pin, with Zn particles as nodes and ZnO bridges as links. The same study reports that bounce height remains at a low plateau early in discharge, then rises sharply at a specific state of charge before settling into a higher plateau. We propose that this abrupt rise corresponds to the moment when the giant component of the Zn–ZnO network first makes mechanical contact with the collector pin.

At low discharge, network clusters and zinc granules remain effectively free-suspended. Upon impact, the rigid shell reverses direction almost instantly while these suspended particles continue downward, and the resulting velocity mismatch is rapidly removed by viscous drag in the gel. Thus, early in discharge, energy is lost both at impact and through this fast internal viscous relaxation.

As discharge proceeds, the largest network cluster becomes a true giant component (as shown in the Supplementary Information) and eventually forms a continuous mechanical path to the collector. Once this happens, most of the anode mass is no longer free-suspended but moves with the rigid body of the cell. The viscous-dissipation channel therefore collapses, producing a sudden drop in internal energy loss and explaining the observed jump in bounce height.

Beyond this transition point, the dynamics are fully described by the model in the previous section. Below the transition, the post-impact velocity can be obtained from momentum and energy conservation:

$$v = (1 - X)\sqrt{v_1^2 - \frac{2\Delta E}{m}} - Xv_1,$$

where v_1 is the pre-impact velocity, m the cell mass, ΔE the impact-related energy loss, and X an effective mass fraction representing Zn/ZnO still free in the gel (with small corrections for unmodeled internal processes). Substituting $\Delta E = \alpha m v_1^3$ from section IIB gives

$$e = \frac{|v|}{|v_1|} = (1 - X)\sqrt{1 - 2\alpha v_1} - X.$$

For small enough α and X, this reduces to $e_{\text{eff}} \approx 1 - \alpha v_1$ as before. A recursion for bounce heights can be shown to follow:

$$h_{n+1} = \left[(1 - X)\sqrt{1 - 2\alpha\sqrt{2gh_n}} - X \right]^2 h_n.$$

where n is the bounce number. Fitting this recursion to measured h_n by optimizing MSME yields an excellent match (Fig. 7), hence supporting the model. For post-transition charge states, the simpler recursion from the previous section applies.

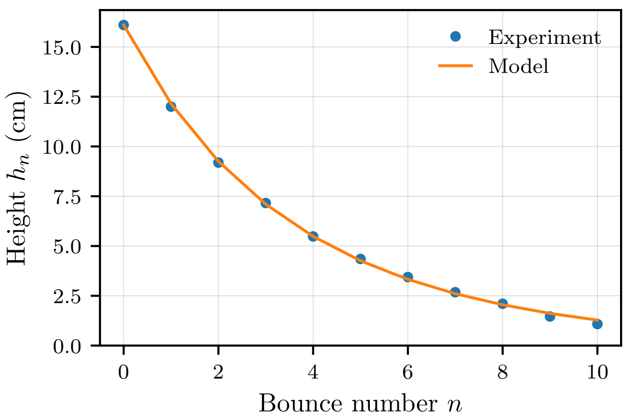

FIG. 7: h_n vs n for 100% SOC on **RS**

III. NON-INVASIVE METHODS

A. Ultrasonic Testing for SOC Estimation

In this study, we employ ultrasonic testing (UT) as a mechanical non-destructive evaluation (NDE) method for analysing battery behaviour. Prior work has demonstrated the usefulness of ultrasound for SOC estimation. For example, Zhang et al. [9] used through-transmission UT to monitor SOC in lithium-ion batteries. Similarly, Bhadra et al. [10] applied pulse-echo ultrasound to track anode density changes during zinc discharge in AA alkaline batteries. Their findings directly motivate and support our approach.

1. *Experimental Setup*

In our experimental setup (Fig. 8), we use 5 MHz sound waves to measure time-of-flight (TOF) and detect internal flaws. The transducers generate vibrations in the material and record both reflected and transmitted signals. Of the two standard ultrasonic techniques: Pulse-Echo (a single transducer emits and receives) and Through-Transmission (two transducers, one sending and one receiving), our experiment employs the **Pulse-Echo technique** due to convenience and better quality of data.

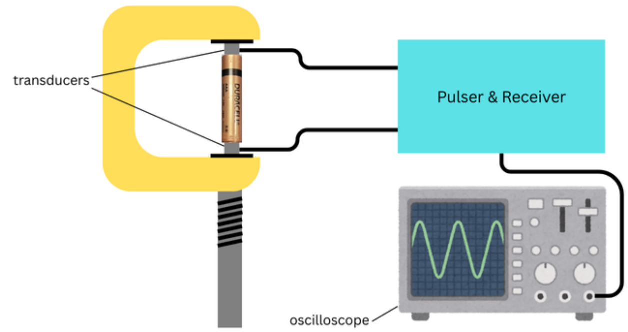

FIG. 8: Experimental Apparatus for UT

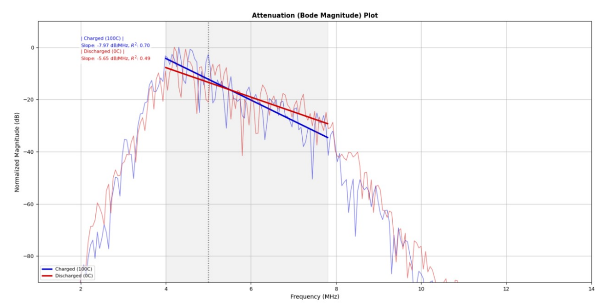

FIG. 9: Frequency-domain Bode plot showing attenuation slope and center frequency shift between charged (100C) and discharged (0C) states.

The TOF measurement gives the sound velocity v. In homogeneous, isotropic materials, the Young's modulus is

$$E = \rho v_L^2 \frac{(1 + \nu)(1 - 2\nu)}{(1 - \nu)},$$

where ρ is the density, v_L the longitudinal wave velocity, and ν Poisson's ratio. It must be mentioned that in batteries, inhomogeneities arising from their multilayer structure and the structural changes during charge-discharge cycles are expected to alter the TOF measurably.

2. *Signal Analysis*

The raw data has been processed by first filtered using a fourth order bandpass filter (with a bandwidth of 4.0 - 7.8 MHz) to isolate the pulse. This cleaned signal was then processed via time-gating to isolate the primary echoes, allowing the extraction of the crucial time-domain metrics: TOF and peak amplitude. The gated signal was transformed into the frequency domain using a 3009 point-FFT to determine the measured center frequency (F_c). Finally, the spectral data was converted to a Bode magnitude plot, and regression was performed to quantify the material's energy absorption and scattering, yielding the attenuation slope (dB/MHz) and the regression fit quality (R^2).

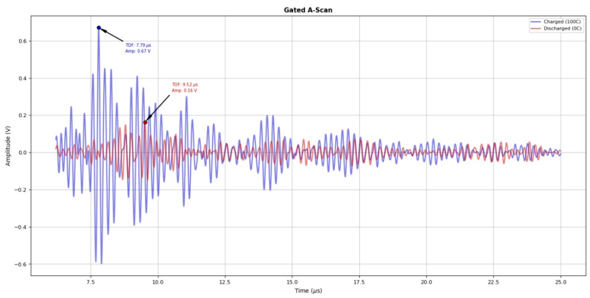

FIG. 10: Gated A-scan ultrasonic signal from the Zn–Mn cell surface. The gated window isolates the primary reflection for time-of-flight (TOF) and amplitude analysis.

3. Results

Using the Pulse Echo Ultrasonic Testing (PEUT), we successfully distinguished between the fully Charged (**100C**) and fully Discharged (**0C**) states of the Zn–Mn cell based on three primary acoustic metrics: Time-of-Flight (TOF), Peak Amplitude, and Acoustic Attenuation.

TABLE II: Time-domain acoustic parameters of charged and discharged Zn–Mn cells.

Acoustic Metric	Charged (100C)	Discharged (0C)
Primary TOF (μs)	7.79	9.53
Peak Amplitude (V)	0.67	0.16

Table II indicates that the discharging process results in an internal medium with a higher acoustic velocity and significantly higher acoustic impedance.

TABLE III: Frequency-domain acoustic parameters of charged and discharged Zn–Mn cells.

Acoustic Metric	Charged (100C)	Discharged (0C)
Attenuation Slope (dB/MHz)	-7.97	-5.65
Linear Fit R^2	0.70	0.49
Center Frequency (MHz)	4.31	4.20

The charged (100C) state showed a different attenuation Slope (**-7.97 dB/MHz**) than the discharged state (**-5.65 dB/MHz**) across the 4.0–7.8 MHz fitting band as observed in III. This indicates that the charged state is more attenuative to high-frequency ultrasound. The better linearity (R^2 = **0.7**) suggests a more consistent scattering mechanism when the cell is fully charged.

B. Coin Test

Lastly, another simple home test for distinguishing a discharged battery from a charged one, is striking the battery with a coin. We noticed a clear audible difference in the pitch and damping. This can be explained by the transformation of the electrolyte from gel to porous like upon discharging [11]. The mechanism of dissipation of sound in gel and porous [12][13] like materials as well as studies on the attenuation of sound in ZnO [14] and MnO2 [15], enable us to draw the conclusion that the dissipation of sound in the discharged battery is lesser than in a charged battery. This can be easily observed at home, and is another simple non-invasive mechanical way to detect whether a battery is charged or not.

IV. CONCLUSIONS

In this report, an analytical explanation for impact collisions based on a contact–force model has been developed. The reliability of the battery bounce test as an indicator of state of charge (SOC) was validated experimentally through the observed relationship between the coefficient of restitution and SOC. It was also found that the coefficient of restitution depends strongly on both the surface properties and the impact velocity. Batteries exhibit significantly reduced rebound on harder surfaces, whereas elastic surfaces such as rubber amplify the difference in bounce height between charged and discharged cells. Additionally, the coefficient of restitution decreases with increasing impact velocity and increases as the state of charge decreases. Furthermore, the analysis confirms that during the initial stages of discharge, internal viscous relaxation introduces additional energy dissipation. As a result, the post-impact velocity expression deviates slightly from that of the remaining motion, reflecting the transient internal mechanical losses present early in the discharge process.The preliminary comparative analysis using PEUT demonstrates that acoustic parameters can distinguish between the two extreme states of charge. These findings establish distinct acoustic signatures—TOF, amplitude, and attenuation—providing a foundation for a full SOC-cycle study in future work.

The charged (100C) state showed a different attenuation slope (**-11.875 dB/MHz**) than the discharged state (**-10.980 dB/MHz**) across the 4.0–7.8 MHz fitting band. This indicates that the charged state is more attenuative to high-frequency ultrasound. The better linearity (R^2 = **0.599**) suggests a more consistent scattering mechanism when the cell is fully charged.

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